

MAT 2305- Real Analysis II
Assignment 03

Answers all the questions and submit solution for **1-6 , 9** and **11** on or before 13th January 2025

1. Compute the lower sum, $L(P, f)$, and the upper sum, $U(P, f)$, of the function, f , in each of the following cases, where P , is the selected partition of the interval $[0, 1]$:
 - i. $f(x) = x^2$ on $[0, 1]$ and $P = \left\{0, \frac{1}{4}, \frac{2}{4}, \frac{3}{4}, 1\right\}$;
 - ii. $f(x) = x$ on $[0, 1]$ and $P = \left\{0, \frac{1}{3}, \frac{2}{3}, 1\right\}$.
2. Let $f: [a, b] \rightarrow \mathbb{R}$ be function given by $f(x) = k$, where k is a constant. Show that f is Riemann integrable on $[a, b]$.
3. Show that the function $f(x) = x^3$ defined on $[0, a]$; $a > 0$, is Riemann integrable on $[0, a]$ and that $\int_0^a f(x)dx = \frac{a^4}{4}$.
4. Show that $f(x) = \sin x$ is Riemann integrable on $\left[0, \frac{\pi}{2}\right]$.
5. Let $f: [0, 2] \rightarrow \mathbb{R}$ be a function defined by

$$f(x) = \begin{cases} x^2 + x & \text{if } x \text{ is rational,} \\ x^2 + x^3 & \text{if } x \text{ is irrational.} \end{cases}$$

Evaluate the upper and lower Riemann integrals of f over $[0, 2]$ and show that f is not Riemann integrable on $[0, 2]$.

6. Is the following function Riemann integrable over $[0, 2]$?

$$f(x) = \begin{cases} 1 & \text{if } x \text{ is rational,} \\ 0 & \text{if } x \text{ is irrational.} \end{cases}$$

7. Let $0 < a < b$ and

$$f(x) = \begin{cases} 1 & \text{if } x \text{ is rational,} \\ 0 & \text{if } x \text{ is irrational.} \end{cases}$$

Find the upper and lower Riemann integrals of $f(x)$ over $[a, b]$.

8. Let $0 < a < b$ and

$$f(x) = \begin{cases} 0 & \text{if } x \text{ is rational,} \\ x & \text{if } x \text{ is irrational.} \end{cases}$$

Find the upper and lower Riemann integrals of $f(x)$ over $[a, b]$ and determine whether f is Riemann integrable over $[a, b]$.

9. Let $0 < a < b$ and

$$f(x) = \begin{cases} \frac{1}{q} & \text{if } x \text{ is rational and } x = \frac{p}{q}, \text{ where } p \text{ and } q \text{ are coprime} \\ 0 & \text{if } x \text{ is irrational} \end{cases}$$

Show that f is Riemann integrable over $[a, b]$ and evaluate $\int_a^b f(x)dx$.

10. Let $f: [a, b] \rightarrow \mathbb{R}$ be a Riemann integrable function.
Show that $|f(x)|$ is Riemann integrable over $[a, b]$ and

$$\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx.$$

11. Evaluate the following improper integrals:

i. $\int_1^{\infty} \frac{1+x}{x^3} dx$

ii. $\int_{-\infty}^{\infty} \frac{1}{1+x^2} dx$

iii. $\int_1^{\infty} \frac{1+x}{x^3} dx$

iv. $\int_0^{\pi/2} \frac{\cos x}{\sqrt[3]{\sin x}} dx$

v. $\int_0^{\infty} \sqrt{x} e^{-\sqrt{x}} dx$